

```
df.rename(columns = d, inplace = False)

abc=pd.DataFrame(df['key'].str.isdecimal())
abc.head()
list1=abc["key"]
df["deimalornot_key"]=list1
abc=pd.DataFrame(df['key'].str.contains("/", case=False))
abc.head()
list1=abc["key"]
df["key_has_"]=list1
abc=pd.DataFrame(df['key'].str.contains(" ", case=False))
abc.head()
list1=abc["key"]
df["space_in_key"]=list1
df["noofchar_value"]=df["value"].str.len()
df.head()
```

Now that the features are extracted and built, the next step is to train the model. We are using the Sklearn library for the machine learning models. The following lines are used for assigning the features:

```
cols=["noofcharkey", "key_has_@", "deimalornot_key", "key_
has_", "space_in_key"]
```

```
X = df[cols] # Features
y = df["label"] # Target variable
```

For this particular use case, after trying all the algorithms, I have concluded that random forests and decision tree algorithms give close to 94% accuracy and hence the best results. The following code is used to train the model using the given dataset and algorithm, as shown in Figure 5.

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size = 0.20, random_state = 99)
from sklearn.ensemble import RandomForestClassifier
model = RandomForestClassifier()
model.fit(X_train, y_train)
model.score(X_test, y_test)
```

We can save this model as a pickle file or a *.pmml* file and use it in Python or Java codes, respectively. For that, use the following code:

```
from sklearn2pmml import sklearn2pmml
sklearn2pmml(pipeline, "regression_model.pmml", with_repr =
True)
```

|       | key                        | value         | label   | final          | noofcharkey | key_has_@ | space_in_value | deimalornot_key | key_has_/ | space_in_key | noofchar_value |
|-------|----------------------------|---------------|---------|----------------|-------------|-----------|----------------|-----------------|-----------|--------------|----------------|
| 8310  | 33517752                   | accountnumber | pil     | account number | 8           | False     | False          | True            | False     | False        | 13             |
| 7442  | 2123 2661 9525 0328        | cc            | pil     | credit card    | 19          | False     | False          | False           | False     | True         | 2              |
| 4703  | 10/03/25                   | birthday      | pil     | birthday       | 8           | False     | False          | False           | True      | False        | 8              |
| 8482  | 36824119                   | bankaccount   | pil     | account number | 8           | False     | False          | True            | False     | False        | 11             |
| 9904  | 7158319164                 | bankaccount   | pil     | account number | 10          | False     | False          | True            | False     | False        | 11             |
| ...   | ...                        | ...           | ...     | ...            | ...         | ...       | ...            | ...             | ...       | ...          | ...            |
| 2164  | 27623933661                | mobile        | pil     | phone number   | 11          | False     | False          | True            | False     | False        | 6              |
| 25    | Ernest                     | name          | pil     | name           | 6           | False     | False          | False           | False     | False        | 4              |
| 9431  | 17170556                   | NEFT          | pil     | account number | 8           | False     | False          | True            | False     | False        | 4              |
| 12620 | white                      | race          | Not-pil | Not-pil        | 5           | False     | False          | False           | False     | False        | 4              |
| 3551  | CNBCBrandStudio@nbcuni.com | email         | pil     | email          | 26          | True      | False          | False           | False     | False        | 5              |

13031 rows x 11 columns

Figure 4: The new look of the dataset

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.20, random_state = 99)
from sklearn.ensemble import RandomForestClassifier
model = RandomForestClassifier()
model.fit(X_train, y_train)
model.score(X_test, y_test)
```

0.9428461833525125

Figure 5: Fitting and scoring the model